

Create the Experimental Design in a .txt format

Main Information needed for the experimental .txt file.

The file should contain **4 columns** with the following names:

"Name", "Experiment", "Groups", and "File"

Each column represent the following information:

- Name: the name of the column with the information about each peptide/protein as it is from the given file (except for the MaxQuant file were represents the name you will like to give for that condition).
- Experiment: unique name of each measured sample
- Groups: unique name of the condition of the file
- File: defines which samples **would be consider in the analysis ("T")**, **which not (empty cell)** and **which are the control ("C")** by using T, C, or empty cell.

Note:

You don't need to follow the same order as the column in the input file, you can put the entries in the order that you will like to get the output (plots and tables). SafeQuant will follow the order of the experimental design for all the measurements minus the control, which will be place first.

More columns with additional information can been include in the file but they will not been used in the SafeQuant.

Experimental Design for the Progenesis files:

If the columns with the information have the following names from a Progenesis files:

AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE
Raw abundance													
A			B			C			D			E	
Measure_1	Measure_2	Measure_3	Measure_4	Measure_5	Measure_6	Measure_7	Measure_8	Measure_9	Measure_10	Measure_11	Measure_12	Measure_13	Measure_14

Then the experimental design should have the following format:

Name	Experiment	Groups	File
Measure_1	Sample_1	Control	C
Measure_2	Sample_2	Control	C
Measure_3	Sample_3	Control	C
Measure_4	Sample_4	Condition_1	T
Measure_5	Sample_5	Condition_1	T
Measure_6	Sample_6	Condition_1	T
Measure_7	Sample_7	Condition_2	T
Measure_8	Sample_8	Condition_2	T
Measure_9	Sample_9	Condition_2	T
Measure_10	Sample_10	Condition_3	T
Measure_11	Sample_11	Condition_3	T
Measure_12	Sample_12	Condition_3	T
Measure_13	Sample_13	Condition_4	T
Measure_14	Sample_14	Condition_4	T

Experimental Design for the MaxQuant files:

If the columns inside the MaxQuant output have the following names:

CL	CM	CN
LFQ intensity AD4553 Secreting medium-2	LFQ intensity AD4553 in Non-Secreting medium-1	LFQ intensity AD4553 Non-Secreting medium-2

Then the experimental design should be set as follow:

	A	B	C	D
1	Name	Experiment	Groups	File
2	Measurement 1	AD4553 Secreting medium-1	Condition 2	T
3	Measurement 2	AD4553 Secreting medium-2	Condition 2	T
4	Measurement 3	AD4553 Secreting medium-3	Condition 2	T
5	Measurement 4	AD4553 in Non-Secreting medium-1	Condition 1	T
6	Measurement 5	AD4553 Non-Secreting medium-2	Condition 1	T
7	Measurement 6	AD4553 Non-Secreting medium-3	Condition 1	T
8	Measurement 7	WT + pSW039 Secreting medium-1	Control	C
9	Measurement 8	WT + pSW039 Secreting medium-2	Control	C
10	Measurement 9	WT + pSW039 Secreting medium-3	Control	C

Experimental Design for the Spectronant files:

When the columns of the Spectronant output have the names as follow:

AD	AE	AF	AG	
[8] Measu	[1] Measurement-1.raw.PG.MS2Quantity	[2] Measurement-2.raw.PG.MS2Quantity	[3] Measurement-3.raw.PG.MS2Quantity	[4] Measur

The experimental design should be set as:

Name	Experiment	Groups	File
[1] Measurement-1.raw.PG.MS2Quantity	C2_1	C2	T
[2] Measurement-2.raw.PG.MS2Quantity	C2_2	C2	T
[3] Measurement-3.raw.PG.MS2Quantity	C2_3	C2	T
[4] Measurement-4.raw.PG.MS2Quantity	C2_4	C2	T
[5] Measurement-5.raw.PG.MS2Quantity	C1_1	C1	C
[6] Measurement-6.raw.PG.MS2Quantity	C1_2	C1	C
[7] Measurement-7.raw.PG.MS2Quantity	C1_3	C1	C
[8] Measurement-8.raw.PG.MS2Quantity	C1_4	C1	C

Experimental Design for the converter DIANN files:

When the columns of the DIANN converter have the names as follow:

Q	R	S	T	U
NrOfRazorPrecursors.Measurement-4	Quantity.Measurement-1	Quantity.Measurement-2	Quantity.Measurement-3	Quantity.Measurement-4

The experimental design should be set as:

Name	Experiment	Groups	File
Quantity.Measurement-1	C1_1	C1	T
Quantity.Measurement-2	C1_2	C1	T
Quantity.Measurement-3	C2_1	C2	C
Quantity.Measurement-4	C2_2	C2	C